

**Bayesian Deep Learning for Uncertainty-Aware Predictive Maintenance in Sustainable
Manufacturing**

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CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

The manufacturing sector plays a critical role in the global economy, driving innovation, employment, and industrial development. In Africa, and particularly Nigeria, the manufacturing sector has been positioned as a growth engine under initiatives such as the African Union's Agenda 2063 and Nigeria's Economic Recovery and Growth Plan (ERGP). However, despite its potential, manufacturing in Nigeria faces significant challenges such as high energy consumption, unplanned equipment downtime, resource inefficiency, and heavy environmental footprints.

Nigeria's manufacturing sector is energy intensive. Industries such as cement production (Dangote, Lafarge), oil refining (NNPC refineries), and food and beverage processing consume large amounts of power, often relying on diesel generators due to unstable electricity supply. This not only increases costs but also contributes to greenhouse gas emissions. Unplanned machine failures further compound these problems, leading to production stoppages, material wastage, and inefficient energy use.

Globally, manufacturing is undergoing transformation under Industry 4.0, where data-driven technologies are revolutionizing operations. One key strategy is Predictive Maintenance (PdM), which uses sensor data from equipment (e.g., vibration, temperature, and pressure) to forecast potential failures and estimate the Remaining Useful Life (RUL) of machines. PdM helps industries schedule interventions at the right time, minimizing downtime, avoiding catastrophic failures, and conserving energy.

Traditional PdM approaches in Nigeria still rely on either reactive maintenance (fixing machines only after failure) or preventive maintenance (servicing at fixed intervals). These methods are inefficient because they either risk unplanned downtime or lead to unnecessary repairs.

With the increasing availability of sensor-enabled equipment in Nigerian industries, data-driven PdM presents an opportunity for smarter, more sustainable operations.

Conventional deep learning (DL) techniques have achieved success in RUL estimation. However, they provide point predictions without accounting for uncertainty. This is problematic in noisy, unstable environments such as Nigeria, where sensor data may be incomplete, inconsistent, or disrupted by power fluctuations. Overconfident but inaccurate predictions can cause poor maintenance decisions, leading to wasted resources and energy.

Bayesian Deep Learning (BDL) overcomes these limitations by combining the power of deep neural networks with Bayesian probability theory. BDL not only predicts equipment failure but also provides confidence intervals that capture uncertainty. For sustainable manufacturing in Nigeria, this approach is valuable because it helps industries plan maintenance more accurately, reduce unnecessary interventions, optimize energy use, and minimize emissions—aligning with both Nigeria’s Energy Transition Plan targeting 2060 for net-zero goals.

This research explores the development of a BDL-based predictive maintenance framework tailored to Nigeria’s manufacturing sector, aiming to integrate sustainability metrics (e.g., energy consumption and emissions) into equipment health prediction.

1.2 Statement of the Problem

Despite advances in deep learning for PdM, existing models remain limited in handling uncertainty, often producing overconfident predictions. This shortcoming undermines their reliability, particularly in sustainability-focused industries where misinformed decisions can lead to increased downtime, higher energy usage, and environmental waste. Recent works such as Della Libera et al. (2024) have introduced Bayesian deep learning for RUL prediction via Stein Variational Gradient Descent, while Thompson Sampling has also been explored for uncertainty

quantification in PdM (Li et al., 2021). However, these approaches primarily optimize prediction accuracy, with little emphasis on integrating sustainability metrics such as energy efficiency and waste reduction.

In Nigeria and other emerging economies, this gap is more pressing. Manufacturing industries struggle with unreliable energy supply and high resource costs, making sustainability-driven PdM both a necessity and a challenge. There is therefore a need for hybrid frameworks that combine uncertainty-aware modeling with sustainability optimization.

1.3 Aim and Objectives of the Project Work

The main aim of this project is to develop and evaluate a Bayesian Deep Learning framework for predictive maintenance in sustainable manufacturing contexts. The specific objectives are:

- To identify key sources of uncertainty in predictive maintenance data and incorporate Bayesian methods into deep learning models for improved Remaining Useful Life (RUL) estimation.
- To develop and train BDL models that optimize predictions for both accuracy and sustainability outcomes (e.g., reduced energy waste).
- To compare the performance of BDL models with conventional deep learning baselines in terms of predictive reliability, uncertainty calibration, and sustainability impacts.
- To provide actionable guidelines for deploying BDL-enabled PdM systems in Nigerian and global manufacturing contexts.

1.4 Significance of Project Work

This study contributes to the intersection of **Sustainable AI** and predictive maintenance. First, it addresses the research gap in uncertainty-aware PdM by adapting Bayesian Deep Learning

techniques for real-world industrial data. Second, it introduces sustainability metrics into the evaluation framework, which is largely absent in prior PdM studies. Third, for Nigeria and other emerging economies, this project offers practical pathways for industries to enhance energy efficiency, reduce carbon footprints, and align with global commitments such as the UN Sustainable Development Goals.

1.5 Scope of the Project Work

The project will focus on applying Bayesian Deep Learning methods to publicly available PdM datasets, such as the NASA C-MAPSS turbofan engine dataset and the UCI CBM dataset. Emphasis will be placed on quantifying predictive uncertainty and evaluating energy-efficiency impacts through synthetic sustainability variables. While the case study focus is on Nigeria's manufacturing sector, the framework will be generalizable to other industrial contexts worldwide.

1.6 Limitations of the Project Work

The study will be limited by the availability of labeled sustainability data in PdM contexts, requiring the use of proxies or synthetic datasets for certain analyses. Computational complexity associated with Bayesian inference may also restrict the scale of experiments, though methods such as Monte Carlo Dropout and Variational Inference will be employed to mitigate these challenges.

1.7 Organisation of the Study

The thesis is structured as follows:

- **Chapter One:** Introduction, providing background, problem statement, objectives, and significance.

- **Chapter Two:** Literature review on predictive maintenance, Bayesian deep learning, and sustainable manufacturing.
- **Chapter Three:** Methodology, outlining data collection, preprocessing, model development, and evaluation methods.
- **Chapter Four:** Results and analysis of model performance.
- **Chapter Five:** Discussion of findings, linking results to research objectives and literature.
- **Chapter Six:** Conclusion and recommendations for research and practice.

1.8 Definition of Terms

- **Predictive Maintenance (PdM):** A data-driven maintenance strategy that anticipates equipment failures before they occur.
- **Remaining Useful Life (RUL):** The estimated time before a machine or component fails.
- **Deep Learning (DL):** A subset of machine learning that uses neural networks with multiple layers to extract patterns from data.
- **Bayesian Deep Learning (BDL):** A fusion of deep learning and Bayesian inference that quantifies uncertainties in model predictions.
- **Sustainable Manufacturing:** Manufacturing processes designed to minimize environmental impacts while maintaining efficiency and productivity.
- **Uncertainty Quantification:** The process of measuring confidence in predictions by modeling sources of variability in data and models.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews key concepts and scholarly works related to predictive maintenance (PdM), Bayesian deep learning (BDL), and sustainable manufacturing. It establishes the theoretical framework of this study, highlights emerging trends in the field, and situates the research within the Nigerian context. The review is organized into six sections: predictive maintenance in manufacturing, Bayesian deep learning in PdM, sustainability in manufacturing, theoretical framework, emerging trends (2024–2025), and applications in Nigeria., identifying gaps that this thesis addresses.

2.2 Predictive Maintenance in Manufacturing

Predictive maintenance has evolved from traditional strategies such as corrective maintenance (repair after breakdown) and preventive maintenance (scheduled servicing) to data-driven Predictive Maintenance, which relies on real-time sensor data. PdM allows industries to estimate the Remaining Useful Life (RUL) of machinery and schedule interventions at the most cost- and energy-efficient time (Carvalho et al., 2019).

Deep learning methods, such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models, have improved RUL prediction accuracy by capturing complex, nonlinear degradation patterns in sensor data (Zhang et al., 2020). However, deterministic DL models often fail in noisy industrial environments, producing overconfident predictions that lead to suboptimal maintenance decisions, increased energy use, and unplanned downtime costing industries billions annually (Susto et al., 2015). This limitation underscores the need for uncertainty-aware models, particularly in sustainable manufacturing, where resource efficiency is critical (Li et al., 2020).

2.3 Bayesian Deep Learning in Predictive Maintenance

Bayesian Deep Learning extends deep learning by incorporating probabilistic reasoning, enabling models to quantify uncertainty in predictions. This is critical for high-stakes applications such as manufacturing, where misinformed decisions can result in costly downtime or wasted energy.

2.3.1 Key Studies

Several studies have applied BDL to PdM:

- **Peng et al. (2020):** Applied Bayes by Backprop (BBB) and Monte Carlo Dropout to train Bayesian CNNs and LSTMs on CMAPSS and IEEE PHM 2012 datasets, achieving superior RUL accuracy (e.g., MAE reduction by 10%) compared to frequentist models. Focused on uncertainty but ignored sustainability metrics.
- **Li et al. (2021):** Modeled RUL as normal, logistic, and Weibull distributions using Bayesian Gated Recurrent Units (GRUs) with Monte Carlo Dropout, improving accuracy via sequential boosting on circuit breaker data. Sustainability was not addressed.
- **Benker et al. (2021):** Combined Hamiltonian Monte Carlo (HMC) and BBB for BDL on CMAPSS, reducing RMSE by 15% through uncertainty-informed predictions. High computational costs, limited scalability, and sustainability was absent.
- **Kraus & Feuerriegel (2019):** Developed interpretable BDL models using variational inference on CMAPSS, emphasizing trust in predictions via confidence intervals. Did not integrate energy or emissions goals.
- **Della Libera et al. (2024):** Applied Stein Variational Gradient Descent (SVGD) for efficient uncertainty quantification, outperforming BBB in convergence and calibration. Did not integrate green objectives.

- **Khan et al. (2025):** Used Bayesian optimization with deep reinforcement learning for energy-efficient PdM in IoT systems, achieving 25% energy savings. Lacked full BDL frameworks and uncertainty quantification.

Table 2.0 Comparison of Relevant Works in Bayesian Deep Learning for Predictive Maintenance

Study	Focus	Methodology	What They Did	What They Missed
Peng et al. (2020)	Uncertainty modeling	Bayesian Backpropagation (BBB) + Monte Carlo Dropout on CNNs/LSTMs	Quantified epistemic and aleatoric uncertainty, improved RUL accuracy	Focused only on accuracy & uncertainty, no sustainability or energy considerations
Li et al. (2021)	Distribution-based RUL estimation	Bayesian GRUs with Monte Carlo Dropout	Modeled RUL distributions, enhanced accuracy via boosting	Did not integrate energy/resource optimization; limited to global datasets
Benker et al. (2021)	Hybrid Bayesian inference	Combined Hamiltonian Monte Carlo (HMC) with BBB	Reduced prediction error via hybrid uncertainty quantification	Heavy computational cost, no sustainability or energy-aware design
Kraus & Feuerriegel (2019)	Interpretability in PdM	Variational BDL	Ensured trust via interpretable predictions	No environmental goals
Della Libera et al. (2024)	Efficient uncertainty quantification	Stein Variational Gradient Descent (SVGD) for BDL	Outperformed BBB on CMAPSS; faster convergence and better calibrated uncertainty	Purely technical contribution, no sustainability integration
Khan et al. (2025)	Energy-efficient PdM	Bayesian optimization + RL	Achieved 25% energy savings in IoT PdM	Lacked full BDL, uncertainty focus
Anishere.M (2025)	Sustainability + BDL in PdM	BDL with SVGD, sustainability losses on CMAPSS/Hydraulic datasets	Quantify uncertainty, optimize energy/emissions in Nigerian context	—

2.3.2 Narrative Discussion

The evolution of PdM reflects a shift from deterministic DL to BDL, addressing uncertainty for reliable predictions. Peng et al. (2020) and Li et al. (2021) established BDL's superiority in accuracy, while Benker et al. (2021) and Kraus & Feuerriegel (2019) enhanced robustness and trust. Della Libera et al. (2024) advanced efficiency with SVGD, achieving faster convergence and better-calibrated uncertainty (e.g., standard deviation of 0.12 vs. 0.38 for BBB). Khan et al. (2025) introduced energy efficiency but used optimization rather than BDL.

Despite these advances, sustainability remains underexplored in BDL-PdM. Most studies focus on technical performance, neglecting energy waste or emissions, critical for 2025 green mandates (Mo et al., 2021). This thesis addresses this gap by developing BDL models with sustainability-aware loss functions, optimizing RUL predictions for energy efficiency and emissions reduction, tailored to Nigeria's resource-constrained manufacturing sector (Akinlabi & Oladokun, 2022).

2.4 Sustainability in Manufacturing and PdM

Sustainable manufacturing seeks to minimize energy use, waste, and emissions while maintaining productivity (Li et al., 2020). PdM contributes by preventing unnecessary repairs, optimizing machine use, and reducing resource waste. Li et al. (2020) reviewed ML-driven PdM in smart factories, noting up to 30% downtime reduction and energy savings in automotive sectors. Khan et al. (2025) achieved 25% energy savings using Bayesian optimization in IoT-based PdM, demonstrating sustainability potential.

In African contexts, PdM can reduce reliance on diesel generators, a major cost and emissions driver in Nigeria's manufacturing (Akinlabi & Oladokun, 2022). However, these studies

rarely integrate BDL’s uncertainty quantification with explicit sustainability goals, a gap this thesis addresses by embedding energy-efficient objectives into BDL frameworks (Babu et al., 2016).

2.5 Theoretical Framework

This study builds upon three intersecting theoretical pillars:

- **Reliability-Centered Maintenance (RCM)** – which seeks to optimize maintenance decisions for safety and cost-effectiveness.
- **Bayesian Learning Theory** – which models knowledge and uncertainty probabilistically.
- **Sustainable Development Theory** – particularly circular economy principles, emphasizing efficiency and minimal waste.

Together, these provide the foundation for a Bayesian Deep Learning PdM framework that not only predicts failures with uncertainty estimates but also aligns with energy and sustainability objectives.

2.6 Emerging Trends (2024–2025)

Recent research highlights key directions shaping PdM and sustainable AI:

- **Green AI:** Models are being optimized to reduce their own computational energy footprints, not just optimize machine usage.
- **Explainable BDL:** Kraus and Feuerriegel (2019) emphasized interpretability in Bayesian PdM, a trend that continues with newer frameworks incorporating explainability for industrial trust.
- **Synthetic Data for PdM:** With real industrial data often scarce, synthetic data generation using GANs or simulations is being explored for model training (Deng et al., 2024).
- **Integration with Digital Twins:** PdM models are increasingly being deployed within digital twin ecosystems for real-time monitoring of entire production lines (Kusiak, 2023).

- **Regional Customization:** Research is moving toward tailoring PdM models for emerging economies, though Nigeria remains largely underrepresented.

2.7 Applications in Nigeria

Nigeria's manufacturing sector is highly energy intensive, dominated by cement, oil refining, and food processing industries. Many firms still rely on reactive or preventive maintenance, which leads to costly breakdowns, downtime, and excess diesel generator usage.

Research focusing on PdM in Nigeria remains limited, with most studies centering on energy efficiency in industrial operations rather than uncertainty-aware machine learning (Akinlabi & Oladokun, 2022). Adopting BDL for PdM in Nigeria could therefore address both industrial efficiency and national sustainability goals, including the **Energy Transition Plan (2060 net-zero target)**.

2.8 Gaps in Literature

The literature shows:

1. PdM has advanced through machine learning and deep learning but lacks uncertainty quantification.
2. BDL frameworks address uncertainty but neglect sustainability dimensions.
3. Sustainable manufacturing frameworks exist, yet PdM integration remains shallow.
4. Nigerian and African manufacturing contexts remain underexplored in BDL-PdM research.

This research contributes by **integrating sustainability metrics into BDL-based PdM**, contextualized for Nigeria's manufacturing sector.

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